

Addressing modes:

The various formats of specifying the operands are called addressing modes.

Memory Addressing modes:

In this addressing mode, one operand (source or destination) of an instruction must be an address of a memory location, which will be calculated by PA.

PA calculation:

$$\text{PA} = [\text{Segment}] \times 10\text{H} + \text{Offset}$$

Where Segment= contents of CS/DS/SS/ES register.

The Offset =Effective Address (EA) =Contents of 8-bit/16-bit values (Displacement /variable) /IP/SP/BP/BX/SI/DI

Or Offset=Effective Address (EA)= Contents of 8-bit/16-bit values + contents of BX +contents of SI or DI.

Example: Offset or EA= [ARRAY] + [BX] + [SI] = 2AH + 5000H + 2000H = 702AH

There are six types of memory addressing modes:

i. Direct or Displacement Addressing mode: In this mode, the EA is directly given in the instruction.

Example: MOV AX,[0500] ; EA=0500H , AX will get data from 0500H address.

ii. Register Indirect Addressing mode: In this mode, the EA is in SI, DI or BX, which is calculated by PA.

$$\text{PA} = [\text{DS}] \times 10\text{H} + \text{EA}$$

Suppose, [DS]= 2000H and EA=[SI]=1200H

So, PA=21200H

Example: MOV AL,[SI]; Calculated PA=21200H is the address of memory location, now the contents of memory location address (if [21200H]=02H) will be moved into AL. So, this a READ operation.

iii. Based Addressing mode: In this mode, the EA is calculated by adding the contents of BX or BP and a displacement (variable).

$$\text{So, PA} = [\text{DS}] \times 10\text{H} + \text{EA}$$

Where $EA = [Variable] + [BX \text{ or } BP]$

Suppose, $[DS] = 3000H$, $[ARRAY] = 05H$ and $[BX] = 2100H$

So, $PA = [DS] \times 10H + \{[ARRAY] + [BX]\} = 32105H$

Example: `MOV BL, ARRAY[BX]`; Calculated $PA = 32105H$ is the address of memory location, now the contents of memory location address (if $[32105H] = 04H$) will be moved into AL. So, **this a READ operation.**

Example: `MOV ARRAY[BX], AL`; Calculated $PA = 32105H$ is the address of memory location, now the contents of AL (if $[AL] = 3AH$) will be moved into the memory location of address 32105H.

So, **this a WRITE operation.**

iv. Indexed Addressing mode: In this mode, the EA is calculated by adding the contents of SI or DI and a displacement (variable).

So, $PA = [DS] \times 10H + EA$

Where $EA = [Variable] + [SI \text{ or } DI]$

Suppose, $[DS] = 1200H$, $[START] = 03H$ and $[SI] = 0400H$

So, $PA = [DS] \times 10H + \{[START] + [SI]\} = 12403H$

Example: `MOV AL, START[SI]`; Calculated $PA = 12403H$ is the address of memory location, now the contents of memory location address (if $[12403H] = 08H$) will be moved into AL. **So, this a READ operation.**

v. Based Indexed Addressing mode: In this mode, the EA is calculated by adding the contents of BX or BP, SI or DI and a displacement (variable).

So, $PA = [DS] \times 10H + EA$

Where $EA = [Variable] + [BX \text{ or } BP + SI \text{ or } DI]$

Suppose, $[DS] = 8000H$, $[DISPLACEMENT] = 07H$, $[BX] = 1500H$ and $[SI] = 0500H$.

So, $PA = [DS] \times 10H + \{[DISPLACEMENT] + [BX] + [SI]\} = 81A07H$

Example: `MOV CL, DISPLACEMENT[BX][SI]`; Calculated $PA = 81A07H$ is the address of memory location, now the contents of memory location address (if $[81A07H] = 5BH$) will be moved into AL. **So, this a READ operation.**

vi. String Addressing mode: This addressing mode is related to string

instructions. In this mode, the value of SI or DI are incremented or decremented upon the value of direction flag (DF). If the value of DF=0, SI or DI value will be auto incremented. If the value of DF=1, SI or DI value will be auto decremented.

Example: MOVS B ; MOVS Move the string, B Byte
MOVS W ; MOVS Move the string, W Word

****Suppose, [SI]=0400H, [DI]=0500H, [DF]=1, [DS]=3000H, [ES]=5000H, [30400H]=02H, [50500H]=05H. Justify the following instruction: MOVS B.**

Solution: In this addressing mode, the following steps will be followed:

- i. At first, the two physical addresses (PA1 and PA2) will be calculated:
 $PA1 = [DS] \times 10H + [SI]$ and $PA2 = [ES] \times 10H + [DI]$
- ii. Assign the PAs values from the given values.
- iii. The contents of PA1 will be moved into PA2.
- iv. Write the revised values of PA1 and PA2 after the instruction will be executed.
- v. Finally, the revised values of SI and DI will be written according to the given value of DF.

Now, $PA1 = [DS] \times 10H + [SI] = 30400H$
 $PA2 = [ES] \times 10H + [DI] = 50500H$

Before instruction execution,

$PA1 = [30400H] = 02H$ and $PA2 = [50500H] = 05H$

After instruction execution,

$PA1 = [30400H] = 02H$ and $PA2 = [50500H] = 02H$

Before instruction execution,

$[SI] = 0400H$ and $[DI] = 0500H$

After instruction execution,

$[SI] = 0399H$ and $[DI] = 0499H$; Because $DF=1$

Solve the following Home Works (HW):

1. Calculate the Physical Addresses with comments of the following instructions:

i). MOV [BX],AL; ii). MOV BL,[SI]; where [DS]=3000H, [BX]=250AH and [SI]=0300H.

2. Mention the name of addressing modes, find out PAs of the following instructions:

i). MOV AL,START[BX]; ii). MOV BETA[DI],CL and MOV AL,ARRAY[BX][SI]
Where [DS]=2000H, [BX]=2A25H, [SI]=0700H, [DI]=0200H, [START]=02H, [BETA]=03H and [ARRAY]=04H.

3. Justify the below instruction with the name of addressing mode:

i). MOVS B

Assume the following data prior to execution of the above instruction:

[SI]=0300H, [DI]=0400H, [DF]=0, [BX]=5000H, [DS]=2000H, [ES]=3000H, [20300H]=05H, [30400H]=09H.

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